

MORON AB, Spain

WMO: 083970

Lat: 37.164N

Lon: 5.611W

Elev: 285

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.4	35.6	24.9	19.2	43.5	28.7	22.9	43.5	25.6	57.4	22.3	57.7	3.3	90	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	28.1	102.4	73.3	99.8	72.4	96.7	71.4	76.1	96.0	74.5	93.7	73.2	91.8	7.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	112.0	82.4	68.4	105.5	80.5	67.8	103.3	79.8	39.5	96.5	38.1	94.0	36.9	91.9	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	18.3	15.9	DB	29.2	108.1	4.1	3.7	26.3	110.7	23.9	112.9	21.6	115.0	18.7	117.7	(4)
(5)			WB	27.8	79.1	3.9	2.0	25.0	80.5	22.7	81.7	20.5	82.9	17.7	84.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.3	50.8	52.8	57.6	61.4	67.8	75.2	80.5	81.5	75.2	68.7	58.1	52.6	(6)	
	DBStd	11.78	4.67	4.85	4.83	5.13	5.89	5.35	4.88	4.33	4.85	4.93	5.50	5.07	(7)	
	HDD50	111	45	26	5	0	0	0	0	0	0	0	5	30	(8)	
	HDD65	1805	441	343	231	130	39	1	0	0	0	21	215	384	(9)	
	CDD50	5676	69	103	241	343	551	755	944	975	756	579	248	112	(10)	
	CDD65	1894	0	0	3	22	125	306	479	510	306	135	8	0	(11)	
	CDH74	22723	0	2	123	418	1644	3721	6154	6354	3076	1174	55	2	(12)	
	CDH80	12018	0	0	11	104	696	1955	3662	3732	1470	383	5	0	(13)	
(14) Wind	WSAvg	6.3	5.8	6.1	6.4	6.8	6.5	6.6	6.6	6.5	6.2	5.7	5.8	6.0	(14)	
(15) Precipitation	PrecAvg	24.9	3.2	2.6	2.2	2.8	1.7	0.4	0.0	0.1	1.4	3.3	3.3	4.0	(15)	
	PrecMax	48.4	13.5	10.2	9.3	8.5	5.3	2.5	0.2	0.6	4.8	9.6	9.2	17.1	(16)	
	PrecMin	11.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0	(17)	
	PrecStd	8.3	2.8	2.1	2.1	1.9	1.4	0.6	0.0	0.1	1.1	2.2	2.3	4.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.2	72.0	81.7	87.5	96.6	102.5	107.2	105.9	100.3	91.6	79.1	71.5	(19)	
		MCWB	56.8	57.9	62.5	65.3	71.0	72.7	75.1	75.3	72.7	67.4	62.8	58.3	(20)	
	2%	DB	64.5	69.5	77.3	82.9	91.4	98.2	102.6	102.3	95.4	87.9	75.2	66.6	(21)	
		MCWB	56.0	57.8	60.6	63.4	67.6	71.1	73.2	73.7	70.4	66.6	62.0	57.8	(22)	
	5%	DB	62.6	66.3	73.7	78.9	87.7	94.6	100.1	99.0	91.8	84.3	71.8	64.4	(23)	
		MCWB	55.9	56.3	59.1	61.9	66.0	69.9	72.4	72.6	69.8	65.9	61.4	57.6	(24)	
	10%	DB	60.6	63.0	70.0	75.0	83.9	91.2	96.6	96.5	88.6	80.8	68.3	62.4	(25)	
		MCWB	55.0	55.1	57.9	60.7	64.7	69.0	71.7	71.5	68.7	65.2	60.2	57.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	62.1	65.6	68.0	72.7	76.1	78.2	78.7	75.2	72.0	68.9	63.5	(27)	
		MCDB	64.0	67.7	76.2	81.0	90.5	95.3	100.0	100.0	92.3	80.5	71.7	65.6	(28)	
	2%	WB	59.3	60.0	62.7	65.4	69.8	73.5	75.9	76.3	73.1	69.8	65.4	61.4	(29)	
		MCDB	61.9	65.0	72.2	78.3	86.2	92.2	97.2	95.9	89.2	79.2	70.6	63.5	(30)	
	5%	WB	57.5	58.4	60.8	63.6	67.7	71.8	74.2	74.6	71.6	68.4	63.4	59.5	(31)	
		MCDB	60.8	63.4	69.9	75.5	82.8	89.8	95.0	93.5	86.1	77.8	68.8	62.6	(32)	
	10%	WB	55.7	56.6	59.2	61.7	65.8	70.2	72.7	73.4	70.2	67.1	61.6	57.5	(33)	
		MCDB	59.6	61.8	67.2	72.1	80.2	86.6	92.6	91.7	83.6	76.6	67.1	61.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	21.1	22.9	23.0	25.2	27.5	29.4	28.1	24.3	21.7	19.4	18.9	(35)	
		MCDBR	22.5	25.8	30.5	31.0	33.3	33.7	34.7	32.6	30.2	28.6	23.8	20.7	(36)	
	5% WB	MCWBR	15.4	16.0	16.9	15.7	15.0	14.3	14.6	14.1	12.9	13.5	13.7	13.4	(37)	
		MCWBR	16.3	20.2	24.7	26.9	28.9	30.0	31.2	29.3	25.4	21.8	18.2	14.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.320	0.344	0.380	0.380	0.395	0.407	0.391	0.405	0.396	0.373	0.339	0.324	(40)	
		taud	2.481	2.419	2.312	2.334	2.320	2.303	2.336	2.302	2.352	2.427	2.486	2.490	(41)	
	Ebn at Noon		276	281	279	285	282	277	281	274	271	267	266	266	(42)	
		Edh at Noon	26	31	38	39	41	41	40	40	36	31	26	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		823	1081	1446	1800	2167	2462	2514	2211	1705	1218	889	729	(44)	
	RadStd		94	139	152	134	160	98	53	68	102	107	82	75	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112

Nomenclature: See separate page